MILITARY SPECIFICATION

PROJECTILE, 155MM, HE: XM864
LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-P-70507(AR), dated 28 December 1984, and is approved for use within the US Army Armament Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 3

3.3: Delete in its entirety

PAGE 5

4.3.1: Delete the following parts in their entirety

"Expulsion charge assembly 9272019
Expulsion charge assembly 9362556"

Change the following drawing numbers:

"From 9362552 to 12630546
From 9362553 to 9391001
From 9362555 to 9391003"

Add additional first article items:

| | | |
|---|---|---|
| "Cup, Expulsion | 9391057 | 25 |
| Plate, Forward | 9381111 | 10 |
| Washer, Expulsion | 12630555 | 25" |

PAGE 6

Table 1: Delete the following parts in their entirety:

"Expulsion charge assembly (dwg 9272019)
Expulsion charge assembly (dwg 9362556)"

Change the following drawing numbers:

"From 9362552 to 12630546
From 9362553 to 9391001
From 9362555 to 9391003"


DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:36Z
Check the source to verify that this is the current version before use.

PAGE 7

Table 1: Add the following new items:

| | | | |
|---|---|---|---|
| "Washer, Expulsion (Dwg. 1263055) Examination for Defects | 25 | 3.2 | 4.4.2.21 |
| Cup, Expulsion (Dwg. 9391057) Examination for Defects | 25 | 3.2 | 4.4.2.22". |

PAGE 9

4.4.1: Delete Item K and substitute the following:

"K. Expulsion charge container assemblies from one lot interfix number from one manufacturer."

Add the following item:

"O. Expulsion cups from one lot interfix number from one manufacturer."

PAGE 10

4.4.2.1: Delete in its entirety.

PAGE 11

4.4.2.2: Delete in its entirety.

PAGE 12

4.4.2.3: Delete in its entirety.

PAGE 13

4.4.2.4: Delete in its entirety.

PAGE 27

4.4.2.17: Change next drawing number from 9362552 to 12630546 Change next higher assembly drawing number from 9362556 to 9391003



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:36Z
Check the source to verify that this is the current version before use.

PAGE 28

4.4.2.18: Change drawing numbers as follows:

From 9362553 to 9391001 and from 9362555 to 9391003

Add major defects 110 and 111 as follos:

"110 Flatness of seal area 0.40%/3.2/Gage
111 Wall thickness 0.40%/3.2/Gage"

PAGE 29

4.4.2.19: Change drawing number 9362555 to 9391003

PAGE 30

4.4.2.20: Change drawing numbers as follows:

"From 9362555 to 9391003
From 9362556 to 9381131"

Add the following major defects:

"105 Cup or lid damaged 0.40%/3.2/visual
106 Gap between cup and lid 360°, max 0.40%/3.2/gage
107 Total length, max 0.40%/3.2/gage".

PAGE 31

4.4.2.21: Replace existing paragraph with new 4.4.2.21 attached.

PAGE 32

4.4.2.22: Replace existing paragraph with new 4.4.2.22 attached.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:36Z
Check the source to verify that this is the current version before use.

QUALITY CONFORMANCE INSPECTION

MIL-P-70507 (AR)
AMENDMENT 1

## CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | | SHEET OF | | DRAWING NUMBER |
|---|---|---|---|---|---|
| 4.4.2.21 | Washer, Expulsion | | | | 12630555 |
| | | | | | NEXT HIGHER ASSEMBLY |
| | | | | | 9391002 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Outside diameter | | 0.40% | 3.2 | Gage |
| 102 | Inside diameter | | 0.40% | 3.2 | Gage |
| 103 | Thickness | | 0.40% | 3.2 | Gage |
| 104 | Angle of taper (two places 180° apart) | | 0.40% | 3.2 | Gage |
| Minor | | | | | |
| 201 | Poor workmanship | | 0.65% | 3.2 | Visual |

NOTES:

AMSMC Form 1570, 1 Feb 85

Replaces DRSMC-QA (D) Form 160, 1 Aug 83, which may not be used.

Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:36Z
Check the source to verify that this is the current version before use.

## CLASSIFICATION OF DEFECTS & TESTS

| PARAGRAPH | TITLE | | SHEET OF | | DRAWING NUMBER 9391057 |
|---|---|---|---|---|---|
| 4.4.2.22 | Cup, Expulsion | | | | NEXT HIGHER ASSEMBLY 9381131 |
| CATEGORY | EXAMINATION OR TEST | NO. OF SAMPLE UNITS | AQL OR 100% | REQUIREMENT PARAGRAPH | PARAGRAPH REFERENCE /INSPECTION METHOD |
| Critical | None defined | | | | |
| Major | | | | | |
| 101 | Pitch diameter of thread | | 0.40% | 3.2 | Gage |
| 102 | Major diameter of thread | | 0.40% | 3.2 | Gage |
| 103 | True position of largest inside diameter with pitch diameter of threads | | 0.40% | 3.2 | Gage |
| 104 | True position of smallest inside diameter with pitch diameter of thread | | 0.40% | 3.2 | Gage |
| 105 | Minimum inside diameter | | 0.40% | 3.2 | Gage |
| 106 | True position of largest outside diameter with pitch diameter of thread | | 0.40% | 3.2 | Gage |
| 107 | Length of threaded area | | 0.40% | 3.2 | Gage |
| 108 | Hardness test (only applies for alternate material) | 10 | Note a | 3.2 | 4.5.13 |
| 109 | Thickness of bottom | | .40% | 3.2 | Gage |
| 110 | Thickness of sidewall | | .40% | 3.2 | Gage |
| 111 | Overall length | | .40% | 3.2 | Gage |
| 112 | Depth to start of angle | | .40% | 3.2 | Gage |
| 113 | Presence of two slots | | .40% | 3.2 | Visual |
| 114 | Largest outside diameter | | .40% | 3.2 | Gage |
| 115 | Perpendicularity of largest outside diameter | | .40% | 3.2 | Gage |
| 116 | Presence of angle between large and small inside diameters | | .40% | 3.2 | Gage |
| Minor | | | | | |
| 201 | Evidence of poor workmanship | | .65% | 3.4 | Visual |

NOTES: a. Each heat treat batch (see 6.7). If any cup fails to comply with the applicable requirement, the heat treat batch represented by the sample shall be rejected.



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:36Z
Check the source to verify that this is the current version before use.

PAGE 34

4.4.2.24: Major 104:

Change "Concentricity" to "True Position".

PAGE 38

4.5.2: Delete paragraph 4.5.2 but retain 4.5.2.1

PAGE 39

4.5.2.1: Change dwg. "936255" to "9391003"

4.5.3: Delete this paragraph.

PAGE 42

4.5.12: Delete the first part of the first sentence and substitute the following:

"After each sample of lid (dwg. 12630546) and empty cup (dwg. 9391001) have been assembled together for this inspection (dwg. 12630560), the assembly shall..."

Change drawing"(9362556)" to "(12630560)"

Insert new paragraph 4.5.13 as follows:

"4.5.13 Hardness test of cup. The cups selected shall be subjected to the hardness test in accordance with the method and procedures of ASTM E 18-67. The load and penetrator shall be in accordance with the hardness scale specified on the applicable drawing. Three readings shall be taken on each cup on the outside surfaces of the closed and approximately 1/4" from the edge and approximately $120^0$ apart. A suitable fixture shall be utilized to provide support under the surface being tested. Such fixtures shall be approved in accordance with 4.4.4. If the average value of the three readings for any cup is less than the value specified in the applicable drawing, that cup shall have failed the requirement. Equipment used for performing this test shall be approved in accordance with paragraph 4.4.4."



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:36Z
Check the source to verify that this is the current version before use.

Insert new paragraph 6.7 as follows:

"6.7 Heat-treatment batch. A heat-treatment batch is herein defined as cups, of the same type, that have been produced from one lot of material, and heat treated at the same time in the same furnace, quench medium and aging process for each of all phases of the entire heat treatment process. When a continuous-type heat treatment process is used, the quantity of cups processed through the furnace for each four (4) hours production, or portion thereof, shall be considered as a heat treatment batch."

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1320-A894)



Source: https://assist.dla.mil -- Downloaded: 2026-10-23T03:36Z
Check the source to verify that this is the current version before use.